VS-03 Data Editing and Enrichment

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1. Purpose and value created

VS-03 is where the quality of the data is actively improved. Standardized data is rarely clean. It contains errors, inconsistencies, gaps, and values that are not yet coded. VS-03 detects these problems, corrects them, fills the gaps, and derives the variables that analysis needs. Unlike VS-02, it deliberately changes values to do so. What makes this safe is full traceability: every edit, imputation, conversion, and derivation is rule-based, documented with a reason, attributable, and reversible. The value created is **reliability and analytical readiness**. Every estimate, model, and publication downstream rests on the microdata that VS-03 produces.

VS-03 corresponds to the business domain **Editing, derivation, and non-response correction** (see *Steady-State Architecture, Business domains*). This section describes the business layer of that domain for all four process types: Design, Implementation, Execution, and Management.

- **Data quality analyst, subject-matter statistician, or operations staff:** the stages in attribute 3, which describe what happens in each production cycle.
- **Methodologist or statistical designer:** the Design processes in attribute 4.
- **System builder or IT staff:** the Implementation processes in attribute 4.
- **Quality or compliance manager:** the Management processes in attribute 4.
- **Manager who wants the big picture:** attributes 1, 2, and 8. Examples and end-to-end scenarios are part of the SAF Toolkit.

Signature discipline: values change, but never silently

VS-03 is the domain that changes values. The guarantee is not that values stay the same, but that no value changes without a record. For every change, the original value, the rule or decision applied, the reason, the actor, and the time are kept. Imputed values are always flagged, so they can be told apart from observed values. Every quality version can be reproduced from its standardized input and the recorded rules. The invariant is: *no undocumented or irreversible change.*

Work inherited from VS-02

VS-02 deliberately leaves all value-changing work to VS-03. VS-03 receives Standardized Data together with the metadata VS-02 attached: the recorded units, currencies, and reference periods, the definition-difference records, and the list of unmapped values. VS-03 then carries out the operations that VS-02 deferred. These are the interpretive coding of free text, the conversion of units and currencies to a common basis, the alignment of reference periods, the reconciliation of values across sources with different definitions, and the derivation of new variables.

PLACEHOLDER: ArchiMate view VS-03 (a) value stream and capabilities

2. Position and state transition

VS-03 is the third primary value stream. It sits between standardization (VS-02) and estimation (VS-04). It receives Standardized Data when VS-02 publishes the *Standardized Data Registered* event, and it hands over Processed Data to VS-04.

VS-03 takes data from State #2 to State #3. Before VS-03, the data is aligned in structure and coding, but its values are unchanged and may contain errors and gaps. After VS-03, it is a registered, immutable quality version of microdata whose content has been checked, corrected, completed, and enriched. It is linked with other sources where applicable, it carries derived variables, and its lineage is fully documented. The quality gate, the minimum metadata set, and the protection of State #3 are described in *State #3: Processed Data* and are not repeated here.

State #3 is the only steady state that is mandatory for all statistical production. Where a source skips State #2, VS-03 takes it directly from State #1. Processed Data is also a product in its own right. Researchers and other statistical domains reuse it directly, and research datasets for microdata services are prepared from it.

From	To	Gate	Event out
#2 Standardized Data (or #1 Raw Data where #2 is skipped)	#3 Processed Data	Gate of State #3 (content quality, residual rule violations, imputation flags, edit lineage, derived-variable metadata, disclosure risk of linkage, reproducibility)	*Processed Data Registered*

PLACEHOLDER: ArchiMate view VS-03 (c) state handover and events: Standardized Data (#2) in the upstream domain, the editing service at the boundary (error detection, editing, imputation, derivation, quality evaluation), Processed Data (#3) inside the editing domain

3. Stages

The stages describe the value that VS-03 adds in every production cycle. Together they form the execution process type of VS-03. In a mature steady state, the editing rules, the imputation methods, and the derivation specifications have been designed and built in advance (see attribute 4). Execution then focuses on applying them, involving analysts where judgment is needed, and recording every change.

Stage	Name	Value added
VS-03.1	Conduct Error Detection and Quality Diagnostics	It is clear where the data is wrong, incomplete, or inconsistent, and which problems matter
VS-03.2	Perform Data Editing and Correction	Errors are corrected and free text is coded, with every change documented and reversible
VS-03.3	Impute Missing or Non-Response Data	Gaps are filled with the best available estimate, clearly flagged as imputed
VS-03.4	Derive Additional Variables and Indicators	The data carries the converted values, links, and derived variables that analysis needs
VS-03.5	Perform Quality Evaluation and Versioning	A traceable Processed Data version with measured quality is registered (State #3)

Each stage below is described with its activities. The column *SAF process* shows the business process from *Business Processes* that the activity belongs to. The column *Other reference* shows the principles, capabilities, or external frameworks that the activity applies.

VS-03.1 Conduct Error Detection and Quality Diagnostics

Value created: *A clear, prioritized picture of where the data is wrong, incomplete, or inconsistent.*

The stage starts when VS-02 publishes the *Standardized Data Registered* event. The designed detection rules run on the incoming version. They cover consistency between fields and with reference data, value ranges, outliers, duplicates, and completeness. Selective-editing scores are computed, so that effort goes to the errors that materially affect the results. The outcome is a diagnostic report with the location, type, and severity of each suspected error. Each error also gets a disposition: correct automatically, flag for review, or accept. No value is changed in this stage; correction happens in VS-03.2.

Activity	Key outputs	SAF process	Other reference
Run consistency, range, outlier, duplicate, and completeness checks	Error diagnostic report	PRD-5.3 Review and validate	1.3.3 Check content
Compute selective-editing scores and prioritize errors	Prioritized error list	PRD-5.3 Review and validate	
Assign a disposition: automatic correction, review, or accept	Disposition list	PRD-5.3 Review and validate	
Log all detected issues in the edit audit trail	Audit trail entries	OVA-4 Process Data Management	SAF-P10 Security by design

VS-03.2 Perform Data Editing and Correction

Value created: *Errors corrected and inconsistencies resolved, with every change documented and reversible.*

This stage corrects the errors found in VS-03.1. Automatic correction rules are applied to the records that the design marked for automatic treatment. Flagged records go to subject-matter statisticians, who review and correct them. For every change, the original value, the rule or decision, a reason code, the actor, and the time are recorded. The stage also carries out the interpretive coding handed over from VS-02, such as coding free-text occupation to ISCO; cases with low confidence are reviewed. Where linked sources disagree on a value, the designed reconciliation rules are applied. No edit may leave a record that violates the validation rules.

Activity	Key outputs	SAF process	Other reference
Apply automatic corrections according to the designed rules	Corrected records	PRD-5.4 Edit and impute	1.3.4 Adjusting and imputing
Route flagged records to subject-matter review	Reviewed corrections	PRD-5.4 Edit and impute	
Code free-text values interpretively, for example occupation to ISCO	Coded values, confidence and review records	PRD-5.2 Classify and code	
Apply reconciliation rules where linked sources disagree	Reconciled values	PRD-5.1 Integrate data; PRD-5.4 Edit and impute	
Record original value, rule, reason, actor, and time for every change	Edit lineage records	OVA-4 Process Data Management	SAF-P10 Security by design

PLACEHOLDER: VS-03 editing flow: detected errors split by selective-editing score into automatic correction, subject-matter review, and accepted records; free-text values passing interpretive coding with a review loop for low confidence; every change writing original value, rule, reason, actor, and time to the edit audit trail

VS-03.3 Impute Missing or Non-Response Data

Value created: *Complete datasets in which gaps are filled with the best available estimate, clearly flagged as imputed.*

This stage fills missing values and non-response with the designed imputation methods. The method is chosen per variable and pattern of missingness: deterministic rules, donor methods, model-based methods, or external sources such as administrative registers. Acceptance constraints make sure that imputed values still satisfy the edit rules. Every imputed value receives an imputation flag and a method code, so imputed and observed values can always be told apart downstream. The inputs of each imputation are recorded in the lineage. Imputed values never overwrite observed values; they only fill genuine gaps.

Activity	Key outputs	SAF process	Other reference
Select and apply the designed imputation method per variable and missingness pattern	Imputed values	PRD-5.4 Edit and impute	1.3.4 Adjusting and imputing
Enforce acceptance constraints: imputed values satisfy the edit rules	Checked imputations	PRD-5.4 Edit and impute	

Activity	Key outputs	SAF process	Other reference
Flag every imputed value with its method code and record its inputs	Imputation flags and lineage	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Log imputation actions in the edit audit trail	Audit trail entries	OVA-4 Process Data Management	SAF-P10 Security by design

VS-03.4 Derive Additional Variables and Indicators

Value created: *Enriched datasets that carry the converted values, links, and analytical variables that downstream production needs.*

This stage calculates the derived variables and indicators specified in Design. Each derived variable is computed from its documented formula and inputs, receives its metadata and classification, and is linked in the lineage to the inputs it came from. The stage also carries out the conversions that VS-02 deferred. Examples are converting turnover from thousands of euro to euro, or aligning a fiscal-year figure to the calendar year. Each conversion is documented with its rule and rate, so the original value can always be recovered. Where datasets are combined, records are linked on the designed keys, and the linkage is recorded in the lineage.

Activity	Key outputs	SAF process	Other reference
Calculate derived variables and indicators from the designed formulas	Derived variables with metadata	PRD-5.5 Derive new variables and units	1.3.2 Deriving variables and units
Convert units, currencies, and reference periods from the bases recorded by VS-02 Data Standardization and Integration	Converted values with conversion lineage	PRD-5.5 Derive new variables and units	
Link datasets on the designed keys where applicable	Integrated dataset with linkage lineage	PRD-5.1 Integrate data	2.1.4 Statistical register management
Record derivation and conversion formulas and inputs as lineage	Derivation lineage	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification

VS-03.5 Perform Quality Evaluation and Versioning

Value created: *A trustworthy, traceable Processed Data version with measured quality improvement.*

This stage evaluates the result and registers the official quality version. Quality is measured against the baseline of State #2: the improvement in completeness and accuracy, the edit and imputation rates, and the derived variables added. The edit lineage is checked for completeness, so that every change is attributable and reversible. The data must then pass the quality gate of State #3. A persistent, immutable quality version is created with a unique identifier and registered in the data catalog with its lineage, imputation flags, derived-variable metadata, and quality assessment. Access permissions are applied. A correction or redelivery later in the cycle always creates a new version; it never overwrites an earlier one. The stage ends by publishing the *Processed Data Registered* event, which tells VS-04 and other consumers that new processed data is available.

Output: Processed Data (State #3), the official handover artifact to the rest of the statistical value chain. Its structure and minimum metadata set are described in *State #3: Processed Data*.

Activity	Key outputs	SAF process	Other reference
Evaluate quality improvement against the baseline of State #2	Quality evaluation report	PRD-5.8 Finalize data files; OVA-1 Quality Management	2.1.3 Statistical quality management
Verify that the edit lineage is complete and reversible	Lineage completeness check	DLM-2 Promote data through a quality gate	SAF-P10 Security by design
Create an immutable, identified quality version of Processed Data	Versioned Processed Data	PRD-5.8 Finalize data files; DLM-5 Store and version data	CSDA Traceability
Register metadata: edit lineage, imputation flags, derived-variable metadata, quality assessment	Catalog registration	DLM-3 Register and catalog data; OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Apply access permissions according to the designed framework	Access-controlled dataset	DLM-4 Classify and protect data	SAF-P10 Security by design, SAF-P11 Privacy by design
Publish the *Processed Data Registered* event to notify VS-04 Estimation, Analysis, and Statistical Modeling	Event notification	DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

PLACEHOLDER: VS-03 quality version structure: the processed data version linked to its edit lineage, imputation record, derived-variable definitions, linkage lineage, and quality assessment

4. The four process types

VS-03 runs through four process types. They do not form a linear sequence but a governance cycle: each process type produces its own output, and that output feeds the others. Design produces metadata, Implementation produces the execution process, Execution produces the statistical data, and Management produces quality reports that feed back into Design. The generic pattern is described in the Overview of this chapter.

The sub-processes below are specific to VS-03. Their names describe the work in the context of editing and enrichment. Each sub-process refines one or more SAF business processes, which are listed under its heading as *Refines*. This keeps every VS-03 sub-process traceable to the reference process model in *Business Processes*.

Design: provides metadata

The design process type produces the **metadata artifacts** that govern all other process types. In VS-03 these are the editing rules, the imputation methods, the derivation and coding specifications, the linkage rules, and the versioning policy. Without them, editing becomes ad hoc and cannot be traced, and management has no baseline to measure quality improvement against. Design runs when a new product or source is introduced, when a method changes, or when Management feeds back an improvement.

Design sub-process	Refines	Main outputs
Design Error-Detection and Validation Rules	PRD-2.5 Design processing and analysis; OVA-1 Quality Management	Validation rule set, selective-editing strategy, error taxonomy and disposition policy

Design sub-process	Refines	Main outputs
Design Imputation Methodology	PRD-2.5 Design processing and analysis	Imputation methodology, donor and matching specification, imputation flagging standard
Design Derivation and Coding Specifications	PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	Derived-variable catalog, coding methodology, conversion specification
Design Integration and Linkage Rules	PRD-2.5 Design processing and analysis; DLM-4 Classify and protect data	Linkage rule set, reconciliation policy, disclosure-risk assessment design
Design Quality Evaluation, Versioning, and Audit Framework	DLM-1 Plan and design data assets; DLM-2 Promote data through a quality gate; DLM-4 Classify and protect data; OVA-1 Quality Management; OVA-4 Process Data Management	Quality criteria, edit audit-trail specification, versioning policy, access control and reproducibility design

Design Error-Detection and Validation Rules

Refines: PRD-2.5 Design processing and analysis; OVA-1 Quality Management.

Editing starts with a clear definition of what counts as an error. This sub-process sets the validation and detection rules. They cover logical and consistency checks within a record, between fields, and against reference data, as well as ranges, outliers, duplicates, and completeness. A selective-editing strategy is designed, so that effort concentrates on errors that materially affect the results. Score functions and thresholds decide which records are corrected automatically, which go to review, and which are accepted as they are. The team also designs the error taxonomy and the disposition options. Together, these rules make up the VS-03 part of the quality gate of State #3.

Activity	Key outputs	SAF process	Other reference
Define consistency, range, and outlier detection rules	Validation rule set	PRD-2.5 Design processing and analysis	1.3.3 Check content
Define duplicate and completeness detection rules	Validation rule set	PRD-2.5 Design processing and analysis	
Design selective-editing score functions and review thresholds	Selective-editing strategy	PRD-2.5 Design processing and analysis	
Define the error taxonomy and disposition options: automatic correction, review, accept	Error taxonomy and disposition policy	OVA-1 Quality Management	GAMSO Quality Management

Design Imputation Methodology

Refines: PRD-2.5 Design processing and analysis.

This sub-process designs how missing values and non-response are completed. The team selects an imputation method per variable and type of missingness. Options are deterministic rules, donor methods such as hot-deck or nearest neighbor, model-based methods such as regression or ratio imputation, and the use of external or administrative sources. Donor pools, matching variables, and acceptance constraints are defined; an imputed value must still satisfy the edit rules. The design requires an imputation flag and a method code on every imputed value. Imputed and observed values can then always be told apart downstream, which VS-04 needs to estimate variance correctly.

Activity	Key outputs	SAF process	Other reference
Select imputation methods per variable and missingness type	Imputation methodology document	PRD-2.5 Design processing and analysis	1.3.4 Adjusting and imputing
Define donor pools, matching variables, and acceptance constraints	Donor and matching specification	PRD-2.5 Design processing and analysis	
Specify the use of external and administrative sources for imputation	Rules for imputation from external sources	PRD-2.5 Design processing and analysis	SAF-P08 Data is shared property
Require imputation flags and method codes on every imputed value	Imputation flagging standard	PRD-2.5 Design processing and analysis	SAF-P09 No data without metadata and classification

Design Derivation and Coding Specifications

Refines: PRD-2.2 Design variables; PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data.

This sub-process specifies the new variables and indicators to derive, and the interpretive coding handed over from VS-02. Each derived variable is defined by a formula, its input variables, its unit, and the classification it belongs to. For interpretive coding, the design sets the classification, such as ISCO for occupation or COICOP for product names. It also sets the approach: manual, computer-assisted, or assisted by machine learning with confidence thresholds. A review workflow is designed for cases with low confidence. Finally, the team designs how units, currencies, and reference periods are converted, using the bases that VS-02 recorded as metadata.

Activity	Key outputs	SAF process	Other reference
Specify derived variables and indicators: formulas, inputs, units, classification	Derived-variable catalog	PRD-2.2 Design variables	1.3.2 Deriving variables and units
Design the interpretive coding approach and its review workflow	Coding methodology	PRD-2.5 Design processing and analysis; DLM-9 Manage reference and master data	
Design the conversion rules for units, currencies, and reference periods	Conversion specification	PRD-2.5 Design processing and analysis	
Define metadata and classification for every derived variable	Derived-variable metadata standard	PRD-2.2 Design variables	SAF-P09 No data without metadata and classification

Design Integration and Linkage Rules

Refines: PRD-2.5 Design processing and analysis; DLM-4 Classify and protect data.

Where Processed Data is combined with other datasets, this sub-process designs the linkage. It specifies the linkage keys, such as the key of the statistical business register that VS-02 aligned. It defines deterministic and probabilistic matching rules. It sets the rules for resolving conflicts where linked sources disagree on a value. It also sets the reconciliation policy for the definition differences that VS-02 recorded as metadata. Linkage of pseudonymized microdata can increase the risk of re-identification. The design therefore includes an assessment of disclosure risk per linkage, in line with SAF-P11.

Activity	Key outputs	SAF process	Other reference
Define linkage keys and deterministic and probabilistic matching rules	Linkage rule set	PRD-2.5 Design processing and analysis	2.1.4 Statistical register management
Design conflict resolution and value reconciliation across sources	Reconciliation policy	PRD-2.5 Design processing and analysis	
Design the disclosure-risk assessment for linkage and enrichment	Disclosure-risk assessment design	DLM-4 Classify and protect data	SAF-P11 Privacy by design; 3.3.1 Privacy management
Define lineage requirements for linked and reconciled values	Linkage lineage specification	PRD-2.5 Design processing and analysis	CSDA Traceability

Design Quality Evaluation, Versioning, and Audit Framework

Refines: DLM-1 Plan and design data assets; DLM-2 Promote data through a quality gate; DLM-4 Classify and protect data; OVA-1 Quality Management; OVA-4 Process Data Management.

This last design sub-process makes sure that quality is measured and that every change can be traced. It defines the criteria for quality evaluation: the improvement in completeness and accuracy against the baseline of State #2. It defines the requirements of the edit audit trail: the original value, the rule, the reason, the actor, and the time of every change. It defines the versioning policy: how versions are identified, what triggers a new version, and how immutability and retention are handled. Role-based access controls for processed microdata are designed in line with SAF-P10. Reproducibility is designed in from the start: a quality version must be regenerable from its standardized input and the recorded rules.

Activity	Key outputs	SAF process	Other reference
Define quality evaluation criteria against the baseline of State #2	Quality criteria document	OVA-1 Quality Management; DLM-2 Promote data through a quality gate	GAMSO Quality Management
Define the edit audit-trail requirements: original value, rule, reason, actor, time	Edit audit-trail specification	OVA-4 Process Data Management	SAF-P10 Security by design
Define the versioning policy: identification, triggers, immutability, retention	Versioning policy	DLM-1 Plan and design data assets	CSDA Traceability
Design role-based access control and reproducibility for processed microdata	Access control and reproducibility design	DLM-4 Classify and protect data	SAF-P10 Security by design; 3.3.4 Logical identity and access control

PLACEHOLDER: VS-03 design process decomposition: inputs (State #2 metadata, methodology standards, quality requirements, legal and privacy requirements), the five design sub-processes in order, and the design artifact each produces, which together govern Implementation, Execution, and Management

Implementation: provides the execution process

The implementation process type takes the metadata artifacts from Design and **builds, configures, and tests** the editing and enrichment pipeline until it is ready for production. Nothing runs operationally until this process type is complete. Components are reused where possible and built only where needed, in line with SAF-P01. Editing and imputation tools developed in the NSI community are good candidates for reuse.

Implementation sub-process	Refines	Main outputs
Build Error-Detection and Editing Engines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows	Detection module, configured editing engine, edit audit-trail capture
Configure Imputation and Coding Services	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data	Imputation services, coding service and review queue, conversion service
Build Derivation, Integration, and Versioning Pipelines	PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data	Derivation pipeline, integration component, versioning store, events
Test and Validate the Pipeline	PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems	Test reports, edit-impact and reproducibility reports, production-ready pipeline

Build Error-Detection and Editing Engines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows.

This sub-process turns the designed validation and editing rules into operational tools. Detection components are built to run the consistency, range, outlier, duplicate, and completeness checks and to compute the selective-editing scores. The editing engine is built or configured to apply automatic corrections according to the designed rules. It routes flagged records to the review workflow for subject-matter statisticians. For every change, it captures the original value, the rule, a reason code, and the actor. All editing logic is version-controlled, so that every edit can be reproduced and attributed.

Activity	Key outputs	SAF process	Other reference
Build detection components: consistency, range, outlier, duplicate, completeness	Production-ready detection module	PRD-3.2 Reuse or build processing and analysis components	SAF-P01 Actively participate in open-source communities to maintain collective control over statistical processes
Build or configure the editing engine: automatic correction and routing to review	Configured editing engine	PRD-3.2 Reuse or build processing and analysis components	1.3.4 Adjusting and imputing
Configure the capture of edit lineage for every change	Edit audit-trail capture	PRD-3.4 Configure workflows	SAF-P10 Security by design
Put all editing logic under version control	Versioned editing repository	PRD-3.4 Configure workflows	SAF-P10 Security by design

Configure Imputation and Coding Services

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-9 Manage reference and master data.

In this sub-process the designed imputation and coding methods become automated services. Imputation services are configured for each method, with donor pools and matching variables loaded and acceptance constraints enforced. Every imputed value is flagged with its method code. The interpretive coding service is configured for free text, with computer-assisted or machine-learning-assisted coding and the designed confidence thresholds. Cases with low confidence go to a manual review queue. The conversion routines for units, currencies, and reference periods are configured to use the bases recorded by VS-02.

Activity	Key outputs	SAF process	Other reference
Configure imputation services per method, with donor pools and constraints	Configured imputation services	PRD-3.2 Reuse or build processing and analysis components	1.3.4 Adjusting and imputing
Enforce imputation flags and method codes	Flagged imputation outputs	PRD-3.4 Configure workflows	SAF-P09 No data without metadata and classification
Configure the interpretive coding service and its review queue	Coding service and review queue	PRD-3.2 Reuse or build processing and analysis components; DLM-9 Manage reference and master data	
Configure the conversion routines for units, currencies, and reference periods	Conversion service	PRD-3.4 Configure workflows	

Build Derivation, Integration, and Versioning Pipelines

Refines: PRD-3.2 Reuse or build processing and analysis components; PRD-3.4 Configure workflows; DLM-3 Register and catalog data; DLM-5 Store and version data.

This sub-process implements derivation, linkage, and quality versioning. Calculators for derived variables are built from the designed formulas, and each produces its variable together with its metadata and classification. The linkage component is configured with the designed keys and matching rules. It applies the conflict-resolution and reconciliation policy and records linkage lineage. The versioning store is built so that each accepted Processed Data version is immutable, identified, reproducible, and accompanied by its full edit lineage. Event notifications are configured, so that VS-04 and other consumers are alerted when a new quality version is registered.

Activity	Key outputs	SAF process	Other reference
Build calculators for derived variables from the designed formulas	Derivation pipeline	PRD-3.2 Reuse or build processing and analysis components	1.3.2 Deriving variables and units
Configure linkage with conflict resolution and reconciliation	Integration component	PRD-3.2 Reuse or build processing and analysis components	2.1.4 Statistical register management
Build the versioning store: immutable, identified, reproducible	Versioning store	DLM-5 Store and version data	CSDA Traceability
Configure the *Processed Data Registered* event for downstream consumers	Event publication	PRD-3.4 Configure workflows; DLM-3 Register and catalog data	SAF-P07 Implement loosely coupled processes and systems

Test and Validate the Pipeline

Refines: PRD-3.5 Test production systems; PRD-3.6 Test statistical business process; PRD-3.7 Finalize production systems.

Before the pipeline goes into production, it must be thoroughly verified. Technical testing covers all individual components: detectors, the editing engine, imputation and coding services, derivation calculators, the linkage component, and the versioning store. Integration testing checks the end-to-end flow from the *Standardized Data Registered* event to the registration of a quality version. An edit-impact test confirms that edits and imputations move the quality indicators in the intended direction, without over-editing. A reproducibility test confirms that a quality version can be regenerated identically from its input and the recorded rules. Imputation is evaluated against a holdout sample with known values. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	
Test integration: end-to-end flow from standardized intake to versioned registration	Integration test report	PRD-3.5 Test production systems	
Run the edit-impact test: intended quality change, no over-editing	Edit-impact report	PRD-3.6 Test statistical business process	
Run the reproducibility test: regenerate a version from input and rules	Reproducibility report	PRD-3.6 Test statistical business process	CSDA Traceability
Evaluate imputation against a holdout sample with known values	Imputation evaluation report	PRD-3.6 Test statistical business process	
Finalize the production system and document operational procedures	Operational procedures manual, production-ready pipeline	PRD-3.7 Finalize production systems	

PLACEHOLDER: VS-03 implementation pipeline: the five design artifacts feeding the three build sub-processes, followed by technical testing, edit-impact testing, and reproducibility testing, resulting in a production-ready editing pipeline and operational procedures

Execution: provides the statistical data

Execution runs the stages of VS-03 in every production cycle. The stages and their activities are described in attribute 3. A cycle starts when VS-02 publishes the *Standardized Data Registered* event, or when a redelivery or correction requires a new quality version. The stages then run in order, with a loop between review and correction until no rule violations remain. Execution applies the designs and the solution as they are; a new type of error or a method that no longer works goes back to Design. Every run ends in one of three outcomes: registered, registered with documented residual quality issues, or held for further review.

Aspect	VS-03
Trigger	*Standardized Data Registered*; a redelivery or correction of an earlier version
Cadence	Per production cycle; per redelivery

Aspect	VS-03
Performed by	ROL-14 Data quality analyst, ROL-15 Subject-matter statistician, ROL-09 Classification steward, ROL-03 Data steward
Input	Standardized data version (#2) and its metadata, editing rules, imputation methods, derivation specifications
Output	Registered Processed Data quality version (#3), its edit lineage, and its quality evaluation
Control point	Gate of State #3 in VS-03.5 Perform Quality Evaluation and Versioning
Exception path	Unresolved records held for subject-matter review; a new error pattern or failing method back to Design; disclosure risk of linkage above the threshold escalated to ROL-16 Disclosure control specialist

Management: provides quality reports

The management process type runs alongside and after execution. It monitors the performance of the editing pipeline, assesses the quality improvement achieved, makes sure that every change can be audited, and feeds improvements back into Design. This closes the governance cycle.

Management sub-process	Refines	Main outputs
Monitor Process Performance	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	Process performance dashboard, throughput report, paradata records
Assess Data Quality	OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use	Quality improvement report, imputation quality report, trend analysis
Manage Methodology and Version Lifecycle	OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use; DLM-10 Archive and preserve data	Methodology version register, metadata quality report, version store audit
Audit and Compliance Reporting	OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	Edit audit summary, disclosure-risk and methodology compliance reports to VS-07 Governance and Compliance
Continuous Improvement	PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan	Improvement recommendations, improvement tracking log

Monitor Process Performance

Refines: OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs.

The operational performance of VS-03 must be tracked systematically across production cycles. The process owner and the quality manager collect and aggregate the key performance indicators (see attribute 9). They monitor throughput against the processing window. They track paradata: who edited what, when, with which rule or model version, and how many manual interventions took place. This information feeds dashboards that give the process owner a current view of the health of the domain. The approach follows the Quality Management capability of GAMSO.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per production cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Monitor throughput against the processing window	Throughput report	OVA-4 Process Data Management	
Track paradata: operator, timing, rule and model versions, manual interventions	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Assess Data Quality

Refines: OVA-1 Quality Management; DLM-8 Monitor data quality, lineage and use.

Beyond process performance, the quality improvement itself is evaluated against the criteria from *Design Quality Evaluation, Versioning, and Audit Framework*. Accuracy improvement compares error rates before and after editing. Completeness improvement compares missingness before and after imputation. Imputation quality is assessed against a holdout sample or known values. Consistency checks that no rule violations remain, and coherence checks the derived variables. Quality indicators are produced per dataset, per source, and per cycle. Trends are tracked to detect over-editing, deteriorating sources, or imputation models that need recalibration. This follows the Data Quality capability of CSDA and SAF capability 2.1.3 *Statistical quality management*.

Activity	Key outputs	SAF process	Other reference
Evaluate accuracy and completeness improvement against the baseline	Quality improvement report (per cycle)	OVA-1 Quality Management	2.1.3 Statistical quality management
Assess imputation quality against a holdout sample or known values	Imputation quality report	OVA-1 Quality Management	CSDA Data Quality
Check for residual rule violations and over-editing	Consistency and over-editing report	DLM-8 Monitor data quality, lineage and use	2.1.3 Statistical quality management
Identify quality trends across sources, variables, and cycles	Quality trend analysis	DLM-8 Monitor data quality, lineage and use	CSDA Data Quality

Manage Methodology and Version Lifecycle

Refines: OVA-2 Metadata Management; DLM-5 Store and version data; DLM-7 Revise data; DLM-8 Monitor data quality, lineage and use; DLM-10 Archive and preserve data.

The editing methods and the quality versions they produce must stay current, complete, and well governed. This sub-process governs the life cycle of editing rules, imputation models, and derivation specifications. It versions them, manages transitions, and retires rules that are no longer needed. It verifies that every processed dataset carries its full edit lineage, imputation flags, and derived-variable metadata, as a compliance check on SAF-P09. It governs the versioning store: retention, immutability, the handling of revisions, and the reconciliation of catalog entries with stored versions. Because State #3 is usually subject to national archival requirements, the transfer of versions to long-term preservation is also governed here. **[VC19]**

Activity	Key outputs	SAF process	Other reference
Govern the life cycle and versions of editing rules, imputation models, and derivations	Methodology version register	OVA-2 Metadata Management	SAF-P03 Active life cycle management
Verify metadata completeness: edit lineage, imputation flags, derived-variable metadata	Metadata quality report	OVA-2 Metadata Management	SAF-P09 No data without metadata and classification
Govern the versioning store: retention, immutability, revisions, reconciliation	Version store audit	DLM-5 Store and version data; DLM-7 Revise data	CSDA Traceability
Track metadata lineage from standardized to processed data	Metadata lineage audit	DLM-8 Monitor data quality, lineage and use	CSDA Traceability
Transfer quality versions to long-term preservation according to retention rules	Archive transfer record	DLM-10 Archive and preserve data	2.1.1 Data lifecycle management

Audit and Compliance Reporting

Refines: OVA-1 Quality Management; OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use.

This sub-process produces the evidence that VS-03 operates within methodological, legal, and organizational requirements. Edit audit reports show that every value change is attributable and reversible, which supports reproducibility and methodological transparency. Disclosure-risk reporting covers linkage and enrichment. It confirms that linkage of pseudonymized microdata stays within the assessed risk of re-identification, in line with SAF-P11. Methodology compliance reporting confirms that approved editing and imputation methods were applied. Security compliance reporting verifies that the designed access controls and the tamper-evident audit trail were applied. These reports meet the obligation of SAF capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance*.

Activity	Key outputs	SAF process	Other reference
Produce edit audit reports: every change attributable and reversible	Edit audit summary	OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	4.1.5 Accountability
Report on disclosure-risk compliance for linkage and enrichment	Disclosure-risk report	OVA-1 Quality Management	SAF-P11 Privacy by design; 3.3.1 Privacy management
Report on methodology compliance: approved editing and imputation methods	Methodology compliance report	OVA-1 Quality Management	4.1.5 Accountability
Report on security compliance: access controls, integrity of the audit trail	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10 Security by design; 3.3.3 Security incident management

Continuous Improvement

Refines: PRD-8.2 Conduct evaluation; PRD-8.3 Agree an action plan.

This last management sub-process closes the governance cycle by analyzing performance and quality data and driving improvements. Trends across production cycles are examined: are error rates falling, is over-editing decreasing, is imputation quality improving, and is the share of coded free text rising? Improvement opportunities are identified. Examples are refining edit rules that produce false alarms, recalibrating imputation models, automating manual edits that recur, improving the coding service, and tuning selective-editing thresholds. Recurring errors that originate at the source are reported to VS-01, so they can be raised with the data provider. Recommendations are fed back to the design sub-processes, prioritized by impact and feasibility, and tracked through the following cycles. Where a new method is needed, the recommendation goes to VS-08. This is SAF capability 4.1.4 *Improvement management* in practice.

Activity	Key outputs	SAF process	Other reference
Analyze quality trends across production cycles	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities: rule refinement, model recalibration, threshold tuning	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Feed recommendations back to the design sub-processes, to VS-01 Data Acquisition and Ingestion, or to VS-08 Innovation and Methodological Development	Design feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management
Prioritize improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Track the implementation of accepted improvements	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management

PLACEHOLDER: VS-03 management feedback loop: data and paradata from Execution flowing through the five management sub-processes, with dashboards to the process owner, quality reports to the quality manager, compliance reports to VS-07, source issues to VS-01, and improvement recommendations back to Design

Metadata and paradata per process type

In VS-03, metadata is what keeps value changes transparent: every edit, imputation, and derivation must describe itself. The table shows what each process type contributes. Data metadata describes the datasets. Process metadata, also called paradata, describes how the process itself ran: who, when, and with which tools.

Process type	Data metadata produced	Process metadata (paradata) produced
Design	Editing and validation rule sets, imputation methods, derivation formulas, linkage rules, versioning policy	Log of design decisions, methodology version, review and approval records
Implementation	Engine configurations, rule parameters, parameters of imputation models, derivation definitions, versioning templates	Build log, test results, edit-impact test outcomes, deployment approvals

Process type	Data metadata produced	Process metadata (paradata) produced
Execution	Edit lineage per record (original value, rule, reason), imputation flags and methods, derived-variable formulas and inputs, linkage lineage, quality version identifiers	Operator identity, timing, volumes, edit and imputation rates, rule and model versions, records of manual overrides
Management	Quality improvement scores against the baseline, edit and imputation rate reports, lineage assessments, trend analyses	Editing dashboards, quality trends, improvement decisions, audit findings

PLACEHOLDER: VS-03 business object lifecycle: design artifacts govern implementation assets, which produce the pipeline that turns Standardized Data (#2) into Processed Data (#3), generating edit lineage, imputation flags, and audit records that management assesses and feeds back to design

5. Actors and roles

VS-03 is an internal value stream in which methodological judgment plays a large role. The table shows the actors and the reference roles involved. Specialized role names used in an NSI map onto these reference roles (see *Role names in the value streams*); for example, the editing methodologist and the imputation specialist are ROL-07 Methodologist, and the coding specialist is ROL-09 Classification steward. VS-02, which supplies the standardized data, is a neighboring value stream and not an actor; the relation is governed by the handover interface in attribute 8. [**VC20**]

Actor or role	ID	Process type	Responsibility in VS-03 Data Editing and Enrichment
International statistical organization	ACT-07	External	Issues methodological guidance on editing, imputation, and classification (for example Eurostat)
Process owner	ROL-01	Management	Accountable for the end-to-end performance of VS-03 Data Editing and Enrichment
Data owner	ROL-02	Management	Accountable for State #3; decides on access, reuse, and retention
Data steward	ROL-03	Execution	Registers quality versions, maintains catalog entries, applies access controls, resolves gate exceptions
Data custodian	ROL-04	Execution	Operates storage, access control, retention, and transfer to long-term preservation
Statistical designer	ROL-06	Design	Specifies derived variables and indicators, their formulas, and their inputs

Actor or role	ID	Process type	Responsibility in VS-03 Data Editing and Enrichment
Methodologist	ROL-07	Design	Designs error-detection rules, the selective-editing strategy, and the imputation methods
Metadata and data architect	ROL-08	Design	Designs the metadata model for edit lineage, imputation flags, and derived variables
Classification steward	ROL-09	Design, Execution	Designs and carries out the interpretive coding of free text; reviews cases with low confidence
Data engineer	ROL-10	Implementation	Builds and configures the editing, imputation, derivation, and linkage components
Platform engineer	ROL-11	Implementation	Configures editing services, connections to reference and donor data, and access controls
Test manager	ROL-12	Implementation	Plans and carries out technical, edit-impact, and reproducibility testing
Data quality analyst	ROL-14	Execution	Runs error detection, evaluates quality improvement, and logs edit lineage and exceptions
Subject-matter statistician	ROL-15	Execution	Reviews flagged records, decides on corrections, and signs off domain-specific edits
Disclosure control specialist	ROL-16	Design, Management	Assesses the disclosure risk of linkage and enrichment
Quality manager	ROL-18	Management	Monitors key performance indicators, produces quality reports, and drives continuous improvement
Compliance officer	ROL-19	Management	Ensures auditability of edits and compliance with disclosure-risk rules, and produces audit reports

6. Capabilities, processes, and functions

Capability foundations

The table shows which capabilities from the SAF capability model VS-03 relies on, and what role they play. NSIs can use it to map the value stream to their own organization.

Capability group	Capabilities	Role in VS-03 Data Editing and Enrichment
1.1 Statistical design and build	1.1.2 Design statistics product, 1.1.3 Build statistics product	Design the editing and imputation methods and the derivation specifications, and build the editing pipeline
1.3 Processing	1.3.2 Deriving variables and units, 1.3.3 Check content, 1.3.4 Adjusting and imputing	Carry out error detection, editing, imputation, and derivation **[VC21]**
2.1 Data management	2.1.1 Data lifecycle management, 2.1.2 Metadata management, 2.1.3 Statistical quality management, 2.1.4 Statistical register management	Manage quality versions and edit metadata, assess quality improvement, link with statistical registers
3.2 Computerization and automation	3.2.1 Application development, 3.2.2 Functional management	Build and maintain editing, imputation, derivation, and linkage components
3.3 Security and privacy	3.3.1 Privacy management, 3.3.2 Security planning, 3.3.4 Logical identity and access control	Manage the disclosure risk of linkage and secure the edit audit trail
4.1 Strategy and leadership	4.1.1 Strategy and governance, 4.1.2 Policy and planning, 4.1.4 Improvement management, 4.1.5 Accountability	Govern the domain, steward the editing methods, plan improvements, report on auditability and compliance

Stages, processes, and functions

The table shows which SAF business processes realize each stage of VS-03 and which business functions they mainly use. It follows the mapping in *Business Processes, Mapping to value stream stages*. The activity tables in attribute 3 can name additional supporting processes. **[VC25]**

Stage	Realized by processes	Main functions
VS-03.1 Conduct Error Detection and Quality Diagnostics	PRD-5.3 Review and validate	FUN-3.3 Process statistical data
VS-03.2 Perform Data Editing and Correction	PRD-5.2 Classify and code; PRD-5.4 Edit and impute	FUN-3.3 Process statistical data
VS-03.3 Impute Missing or Non-Response Data	PRD-5.4 Edit and impute	FUN-3.3 Process statistical data; FUN-4.1 Manage statistical and analytical methodology
VS-03.4 Derive Additional Variables and Indicators	PRD-5.1 Integrate data; PRD-5.5 Derive new variables and units	FUN-3.3 Process statistical data
VS-03.5 Perform Quality Evaluation and Versioning	PRD-5.8 Finalize data files; DLM-2 Promote data through a quality gate; DLM-3 Register and catalog data	FUN-3.3 Process statistical data; FUN-5.2 Assure and review quality; FUN-7.3 Manage data lifecycle and storage

PLACEHOLDER: ArchiMate view VS-03 (b) stages realized by processes and functions

Framework mappings

This reference table maps each process type of VS-03 to the SAF business processes, the international frameworks (GSBPM, GAMSO, CSDA), and the SAF capability model. It is intended for architects and auditors; the rest of this section does not depend on it.

Process type	SAF business processes	GSBPM	GAMSO	CSDA	SAF capabilities
Design	PRD-2.2, PRD-2.5, DLM-1, DLM-2, DLM-4, DLM-9, OVA-1, OVA-4	Phase 2 Design (2.2, 2.5)	Production (design activities), Capability Management	Data Governance, Metadata Management, Data Quality	1.1.2 Design statistics product, 2.1.2 Metadata management, 2.1.3 Statistical quality management
Implementation	PRD-3.2, PRD-3.4 to PRD-3.7, DLM-3, DLM-5, DLM-9	Phase 3 Build (3.2, 3.4 to 3.7)	Capability Management	Data Processing (set-up), Data Governance (configuration)	1.1.3 Build statistics product, 3.2.1 Application development, 3.2.2 Functional management
Execution	PRD-5.1 to PRD-5.5, PRD-5.8, DLM-2 to DLM-5, OVA-1, OVA-2, OVA-4	Phase 5 Process (5.1 Integrate data, 5.2 Classify and code, 5.3 Review and validate, 5.4 Edit and impute, 5.5 Derive new variables and units, 5.8 Finalize data files)	Production (execution)	Data Processing (execution)	1.3.2 Deriving variables and units, 1.3.3 Check content, 1.3.4 Adjusting and imputing
Management	OVA-1, OVA-2, OVA-4, DLM-5, DLM-7, DLM-8, DLM-10, PRD-8.1 to PRD-8.3	Phase 8 Evaluate; overarching Quality Management and Metadata Management	Corporate Support (Quality, Data, and Metadata Management), Capability Management (improvement)	Data Quality, Data Governance, Traceability	2.1.3 Statistical quality management, 4.1.4 Improvement management, 4.1.5 Accountability

7. Principles

The following principles govern VS-03 most directly.

Principle	Statement	Application in VS-03 Data Editing and Enrichment
SAF-P01	Reuse before open source, open source before buying, buying before building	Editing, imputation, and coding tools are reused across products first. Where new tooling is needed, open-source solutions developed with other NSIs, such as selective-editing engines and imputation libraries, come before purchase or own development. **[VC3]**
SAF-P07	Implement loosely coupled processes and systems	Editing is independent from standardization and estimation. Processed Data (State #3) is the decoupling interface: downstream consumers use edited and enriched microdata without knowing how it was corrected or imputed.
SAF-P08	Data is shared property	Processed data, once registered, is available to all authorized consumers. Editing and enrichment are not repeated per product when the same quality version can serve several purposes. Researchers and other domains reuse #3 directly.
SAF-P09	No data without metadata and classification	Every edit, imputation, and derived variable carries metadata: the rule applied, the original value, the imputation method and flag, the derivation formula, and the classification of derived variables. This is what makes the changes transparent and reproducible.
SAF-P10	Security by design	Access controls, integrity verification, and a tamper-evident edit audit trail are built into the pipeline, so every change to a value can be attributed and traced.
SAF-P11	Privacy by design	VS-03 Data Editing and Enrichment works on pseudonymized microdata. Linkage and enrichment with other datasets raise the risk of re-identification, so every linkage is assessed for disclosure risk and data minimization by design.

i Principles that apply across the process types

Several principles apply to all four process types. SAF-P09 is the backbone of the editing discipline: Design defines the rules and the required edit metadata, Execution records the original value, rule, and reason for every change, and Management assesses edit quality and closes gaps. SAF-P10

governs the tamper-evident edit audit trail across all stages. SAF-P11 applies in particular to *Design Integration and Linkage Rules* and to the stages VS-03.2 and VS-03.4, where linkage can raise the risk of re-identification. SAF-P07 governs both the intake of State #2 from VS-02 and the handover of State #3 to VS-04.

8. Interfaces, services, and events

Boundary interfaces

VS-03 is loosely coupled (SAF-P07) to its neighbors through two boundary interfaces. Specifying each boundary explicitly makes the loose-coupling contract concrete and testable: the handover artifact, the metadata it must carry, the event that triggers it, and the point where responsibility transfers. Each interface exchanges a registered, immutable artifact plus its metadata, never internal working data. Neither side therefore needs knowledge of the other's internal processing.

Boundary	Handed over	Required metadata	Trigger	Responsibility transfer
VS-02 Data Standardization and Integration → VS-03 Data Editing and Enrichment (intake)	Registered standardized data version (#2)	Minimum metadata set of State #2, including the list of unmapped values, the recorded units, currencies, and reference periods, the definition-difference records, and the pseudonymization record	*Standardized Data Registered*	VS-02 Data Standardization and Integration keeps custody of the standardized artifact; VS-03 Data Editing and Enrichment takes responsibility for editing, imputation, derivation, interpretive coding, and conversion on receipt of the event
VS-03 Data Editing and Enrichment → VS-04 Estimation, Analysis, and Statistical Modeling (handover)	Registered Processed Data quality version (#3)	Minimum metadata set of State #3: edit lineage, imputation flags and methods, derived-variable formulas and classification, linkage lineage, quality assessment, quality version identifier, classification	*Processed Data Registered*	VS-03 Data Editing and Enrichment keeps custody of the quality version; VS-04 Estimation, Analysis, and Statistical Modeling takes responsibility for weighting, estimation, and analysis on receipt of the event

Business services

Service	Offered to	Realized in	Interfaces
Error detection service	Editing teams	VS-03.1 Conduct Error Detection and Quality Diagnostics	Diagnostic reports, prioritized error lists

Service	Offered to	Realized in	Interfaces
Editing and correction service	Editing teams, subject-matter statisticians	VS-03.2 Perform Data Editing and Correction	Review workbench for analysts, query interface for the edit audit trail
Imputation service	Editing teams, methodologists	VS-03.3 Impute Missing or Non-Response Data	Imputation services per method
Derivation and integration service	Editing teams, downstream analysts	VS-03.4 Derive Additional Variables and Indicators	Derivation catalog, linkage interface
Processed data provisioning service	VS-04 Estimation, Analysis, and Statistical Modeling, VS-06 Dissemination and Service Delivery (microdata services), other products reusing the data	VS-03.5 Perform Quality Evaluation and Versioning	Event notification (*Processed Data Registered*) and a catalog query interface to find quality versions and their edit lineage
Quality version catalog service	All data consumers in the NSI	Management sub-process *Manage Methodology and Version Lifecycle*	Search and browse interface for quality versions, edit lineage, imputation flags, and derived-variable metadata
Design and methodology service	Methodologists, statistical designers	All Design sub-processes of VS-03 Data Editing and Enrichment	Design documents, methodology advice

Business events

Event	Triggers or marks
Design artifacts published	Start of Implementation after a design change
Pipeline production-ready	End of Implementation; Execution can start
Standardized Data Registered	Start of VS-03.1 Conduct Error Detection and Quality Diagnostics for that version (published by VS-02 Data Standardization and Integration)
Errors detected	Outcome of VS-03.1 Conduct Error Detection and Quality Diagnostics; starts automatic correction and review
Edits adjudicated	Completion of subject-matter review in VS-03.2 Perform Data Editing and Correction
Processed Data Registered	Completion of VS-03.5 Perform Quality Evaluation and Versioning; triggers VS-04 Estimation, Analysis, and Statistical Modeling and other consumers
Improvement recommendations issued	Input for Design from *Continuous Improvement*

PLACEHOLDER: VS-03 event choreography between VS-02, the editing teams, subject-matter statisticians, and process management across Design, Implementation, Execution (detect, correct and review, impute, derive and link, evaluate and register), and Management (key performance indicators, quality improvement, compliance, improvement recommendations)

Interaction with the supporting value streams

Value stream	Provides to VS-03	Receives from VS-03
VS-07 Governance and Compliance	Methodology and privacy policy, approval of methods	Edit audit trails, methodology and disclosure-risk compliance evidence
VS-08 Innovation and Methodological Development	New editing, imputation, and coding methods	Methods that no longer work, improvement needs
VS-09 Infrastructure and Platform Enablement	Editing and imputation services, versioning store, catalog, event interfaces	Capacity and service requirements
VS-10 Customer and Stakeholder Engagement	Requirements of researchers and other users for processed microdata	Information on quality and on the methods applied

9. Key performance indicators

The key performance indicators are collected per production cycle in *Monitor Process Performance*. They feed the dashboard that the process owner reviews.

KPI	Description	Measurement
Edit rate	Share of values changed by editing	Per variable, per cycle
Imputation rate	Share of values imputed	Per variable, per cycle
Manual review rate	Share of records routed to subject-matter review	Per source, per cycle
Over-editing indicator	Share of edits with negligible impact on the results	Per cycle
Coding completion rate	Share of free-text values coded successfully	Per classification, per cycle
Processing time	Time from the *Standardized Data Registered* event to the registration of #3	Per dataset, per cycle

10. Toolkit

Examples from NSI practice, the maturity checklist, and end-to-end scenarios for VS-03 are part of the SAF Toolkit.

Toolkit: SAF-TK-V03 VS-03 Data Editing and Enrichment companion [**VC17**]